



PROGRAMME OVERVIEW & PRODUCT DOCUMENTATION

IoT, Robotics & AI Lab Programme

Comprehensive Hands-On STEM Education for Schools · Grade 4–12

Grade 4–12

50+ Projects

3 Kit Tiers

NEP 2020 Aligned

1-Year Support

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50+ IoT, Robotics & AI Projects

Complete Lab Manuals, Step-by-Step Tutorials, and Student Project Guides included with every programme.

Programme Overview

Section 01 — Why Codeeint Labs?

Codeeint Technologies designs and delivers end-to-end IoT, Robotics, and AI learning programmes for schools across India. Our objective is to bridge the gap between classroom theory and real-world application by equipping students with hands-on, project-based experiences from Grade 4 through Grade 12. Every kit, manual, and training session is purpose-built to develop computational thinking, engineering mindset, and problem-solving skills — aligned with NEP 2020 competency goals.

Curriculum Focus Areas

Section 02 — What Students Learn



Introduction to Micro:Bit and MakeCode Graphical (Block-Based) Programming



Design Thinking and Application-Based Problem Solving



Algorithmic Thinking — Loops, Conditional Statements, Logical Operators



Building Block-Based Robotic Design and Physical Computing



Sensor Integration and Introduction to the Internet of Things (IoT)



Introduction to Automation, Actuators, and Sensor Fusion



Python and JavaScript Programming — Intermediate to Advanced Levels



Artificial Intelligence Fundamentals — Computer Vision & Machine Learning



Object-Oriented Programming (OOP) Concepts — Classes, Methods, Inheritance



Wireless Communication — Bluetooth, Remote Control & WiFi-based Systems



Smart Home, Autonomous Systems and Real-World IoT Applications



Experiential and Project-Based Learning with Complete Student Manuals

Kit Tiers – What We Offer

Section 03 — Three progressive tiers, one unified learning journey

Tier 1 — Smart Robotics Car Kit

GRADE 4–6

WHAT MAKES IT UNIQUE

- ✓ Visual drag-and-drop block coding — zero prior experience needed
- ✓ Fully assembled smart car base — students program, not just build
- ✓ Real sensor integration from Day 1 — IR, Ultrasonic, Light
- ✓ Wireless remote control via Bluetooth Micro:Bit
- ✓ Smooth progression path to Python in later grades
- ✓ Complete printed manual with 10+ guided projects

WHAT STUDENTS BUILD

- ✓ Obstacle Avoidance Car
- ✓ Line-Following Robot
- ✓ Object-Following Car
- ✓ Remote-Controlled Smart Car
- ✓ Maze-Solving Car
- ✓ Snake Game & Ping Pong on OLED

Tier 2 — Modular Building & IoT Sensor Kit

GRADE 6–9

WHAT MAKES IT UNIQUE

- ✓ Plug-and-play modular sensors — expand and reconfigure freely
- ✓ Bridges block coding to Python/JavaScript programming
- ✓ IoT applications with real environment sensors
- ✓ Advanced OOP concepts introduced through practical builds
- ✓ WiFi and app-based control capabilities
- ✓ Supports 20+ distinct project configurations

WHAT STUDENTS BUILD

- ✓ Smart Home Automation System
- ✓ Automatic Door with Sensor Control
- ✓ Temperature & Humidity Monitoring Station
- ✓ Pick and Place Robot
- ✓ Color Sorter (Sensor-Based)
- ✓ App-Controlled Devices via Bluetooth

Tier 3 — Advanced Multi-Configuration Engineering Kit

GRADE 8–12

WHAT MAKES IT UNIQUE

WHAT STUDENTS BUILD

- ✓ 16-in-1 configurations — one kit, endless engineering possibilities
- ✓ Full Python and JavaScript programming at advanced level
- ✓ Structural mechanics + servo + motor + sensor fusion
- ✓ Students design their own projects from manual inspiration
- ✓ OOP, modular code architecture, real-world robotics logic
- ✓ 50+ project ideas documented — student-led exploration encouraged

- ✓ Walking Spider Bot with Remote
- ✓ Bi-Ped Humanoid Robot
- ✓ Catapult with Precision Control
- ✓ Oscillating Fan with Environmental Sensing
- ✓ Autonomous Smart Car with Gyro
- ✓ Student-Invented Original Projects

 **AI Deep-Dive Add-On (Grade 10–12):** For schools that want to go further — advanced AI vision, machine learning models, and edge computing projects using high-performance hardware. Available as an upgrade to any lab setup.

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CURRICULUM MAP

Grade-wise Curriculum Map

Section 04 — Grade 4 to 12 · 30 Hours Theory + 30 Hours Practical per Year

GRADE	KIT TIER	CODING LEVEL	CORE FOCUS THEME	SAMPLE STUDENT PROJECTS
Grade 4	Tier 1	Block-Based (MakeCode)	Intro to Micro:Bit, Sensors & Basic Robotics	Walking Spider, Oscillating Fan, Catapult, Smart Lights
Grade 5	Tier 1	Block + Intro Python/JS	Smart Car Mechanics & Autonomous Movement	Obstacle Avoidance Car, Line Follower, Maze Solver, Remote Car
Grade 6	Tier 2	Python & JavaScript	Sensor Integration & IoT Fundamentals	Emergency Lamp, Range Finder, Pick & Place Robot, Color Sorter
Grade 7	Tier 2	Python & JavaScript + OOP	Smart Automation & App-Controlled Systems	Smart Home (4 levels), Automatic Door, Autonomous Car, Gyro Car
Grade 8	Tier 3	Python + OOP (Advanced)	Multi-Config Engineering & Advanced Robotics	Bi-Ped Robot, Fishing Bot, Motion Detection Systems, Lifting Platform
Grade 9	Tier 3	Python / JS — Advanced	Complex Systems, Sensor Fusion & Autonomous Bots	Walking Spider with Remote, Catapult with Control, Freestyle Robot Design

GRADE	KIT TIER	CODING LEVEL	CORE FOCUS THEME	SAMPLE STUDENT PROJECTS
Grade 10	Tier 3	Python — Project-Led	Self-Directed Engineering & System Design	Student-Designed Projects, Competition-Ready Robots, Integrated IoT Systems
Grade 11	Tier 3 + AI	Python — AI / ML Intro	Machine Learning, Computer Vision Foundations	Object Detection Models, AI-Guided Robots, Visual Classification Projects
Grade 12	Tier 3 + AI	Python — Advanced AI	Edge AI, Autonomous Systems & Innovation Projects	Full AI Capstone Projects, Autonomous Navigation, Industry-Standard Demos

Programming Ladder

STAGE 1 — BLOCKS

- MakeCode Drag-and-Drop
- Visual logic flow
- No syntax required
- Grade 4–5

STAGE 2 — PYTHON & JS

- Real code, real syntax
- Loops, functions, OOP
- Sensor & hardware control
- Grade 6–10

STAGE 3 — AI & EDGE

- Machine Learning models
- Computer Vision
- AI-guided robotics
- Grade 11–12

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LEARNING OUTCOMES

Learning Outcomes

Section 05 — What students will know and be able to do

50+

Hands-On Projects & Experiments

9

Grade Levels Covered (4–12)

3

Kit Tiers + AI Upgrade

60h

Theory + Practical Per Year

TECHNICAL SKILLS DEVELOPED

- Build and program sensor-integrated robots from scratch

FUTURE-READY SKILLS GAINED

- Computational thinking and structured problem solving

- ▶ Control motors, servos, and actuators through code
- ▶ Implement infrared and ultrasonic-based detection systems
- ▶ Develop obstacle avoidance and line-tracking algorithms
- ▶ Create wireless control systems using Bluetooth
- ▶ Program autonomous vehicles with gyroscopic control
- ▶ Design smart home automation systems with app integration
- ▶ Apply Python/JavaScript to solve real engineering problems
- ▶ Introduction to machine learning and AI model deployment

- ▶ Engineering design process — prototype, test, iterate
- ▶ Reading and using technical documentation and manuals
- ▶ Independent project planning and execution
- ▶ Collaboration and peer learning in lab environments
- ▶ Confidence to explore beyond guided instructions
- ▶ Familiarity with industry-standard tools and concepts
- ▶ Foundation for higher education in CS, ECE, and AI fields
- ▶ Competitive readiness for science fairs and olympiads

What **Schools** Gain

NEP 2020 COMPLIANCE

- ▶ Competency-based learning structure
- ▶ Interdisciplinary STEM approach
- ▶ Activity and outcome-focused curriculum

TEACHER EMPOWERMENT

- ▶ Full teacher training included
- ▶ Instructor-ready lesson plans
- ▶ Ongoing support for 12 months

SCHOOL DIFFERENTIATION

- ▶ Visible, active STEM labs
- ▶ Competition-winning student projects
- ▶ Admissions edge and brand value

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SERVICES & RESOURCES

Services & Programmes

Section 06 — Everything you need, managed end to end



LAB SETUP & INSTALLATION

- ▶ Full lab design and layout planning
- ▶ All hardware supplied and installed



YEAR-LONG TRAINING PROGRAMME

- ▶ Structured grade-wise curriculum delivery

- ▶ Kit inventory management and cataloguing
- ▶ Workbench configuration for student groups
- ▶ Safety and storage setup included

- ▶ Certified trainer visits included
- ▶ Monthly session planning and progress tracking
- ▶ 12 months of post-setup teacher support
- ▶ Mid-year and end-year assessments

BOOTCAMPS (2 & 4 DAY)

- ▶ Intensive hands-on project sessions
- ▶ Perfect for events, holidays, and fairs
- ▶ Themed camps — Robotics, IoT, AI Vision
- ▶ Certificate of participation for students
- ▶ Expert trainers provided by Codeeint

HOBBY KITS FOR STUDENTS

- ▶ Take-home kits for independent learning
- ▶ Complete printed manual with project ideas
- ▶ Online session support available
- ▶ Students encouraged to go beyond the manual
- ▶ Ideal for gifted learners and enthusiasts

AI DEEP-DIVE PROGRAMME (GRADE 10–12 ADD-ON)

- ▶ AI Fundamentals and Model Training
- ▶ Real-time AI inference on hardware
- ▶ Edge AI concepts and deployment
- ▶ Computer Vision — Object Detection & Classification
- ▶ Student-led AI capstone projects
- ▶ Industry-standard workflows and documentation

Resources — Responsibilities

Section 07 — What Codeeint brings, and what the school provides

CODEEINT BRINGS

- All hardware kits and components for every grade
- All software, programming environments, and licences
- Printed lab manuals and student project guides (50+)
- Industry-certified trainers for all sessions
- Curriculum design, lesson plans, and assessments
- Online support and helpdesk for teachers
- Bootcamp facilitation and event support
- 12-month post-installation support guarantee

SCHOOL PROVIDES

- Dedicated lab room or classroom space
- Workbench tables for student groups
- Basic computer lab (Intel i3, Windows OS)
- Projector or display screen for theory sessions
- Internet connectivity (optional for most modules)
- Assigned teacher co-ordinator for scheduling

SOFTWARE PACKAGE

50+ IoT and Robotics Projects with complete Lab Manuals, Video Tutorials, and Student Innovation Guides included with every programme at no extra cost.



Get in Touch

Reach out. Request a Demo. We are ready.

REQUEST A FREE DEMO →



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Grade 4–12

50+ Projects

NEP 2020

1-Year Support

Certified Trainers

